1. Weight Initialization for Weight Sharing Mechanism:

in a convolutional layer, the invariant that all neurons in the same feature map have the same set of weights.

intuitively, we can ensure that this invariant still holds true by first making sure that all neurons in the same feature map have the same weight initialization,

and then applying identical updates to all these set of weights that are supposed to be identical to each other.

2. The Max Pooling Operation in Backpropagation:

we also need to address how to use backpropagation with the max pooling operation.

we simply backpropagate an error term only to the neuron providing the corresponding input with the maximum value

because the other neurons providing the corresponding inputs but not the maximum value clearly do not affect the overall error.

gradient descent method: $\vec{W}' = \vec{W} + \eta \cdot (-\nabla e)$

$$\begin{bmatrix} W'_{G0} \\ W'_{G1} \\ W'_{G2} \\ W'_{F0} \\ W'_{F1} \\ W'_{F2} \end{bmatrix} = \begin{bmatrix} W_{G0} \\ W_{G1} \\ W_{G2} \\ W_{F0} \\ W_{F1} \\ W_{F2} \end{bmatrix} + \eta \cdot \begin{bmatrix} E_{ror}G \cdot \text{the input of } W_{G0} \\ E_{ror}G \cdot \text{the input of } W_{G1} \\ E_{ror}G \cdot \text{the input of } W_{G2} \\ E_{ror}F \cdot \text{the input of } W_{F0} \\ E_{ror}F \cdot \text{the input of } W_{F1} \\ E_{ror}F \cdot \text{the input of } W_{F2} \end{bmatrix}$$

3. Weight Adjustment in A Convolutional Layer:

A. each weight in a convolutional layer will appear multiple times in the gradient descent equation,

whereas each weight appears only once in the gradient descent equation for a fully connected layer.

B. the resulting partial derivative with respect to a specific weight turns out to be a sum of the partial derivatives with respect to different inputs scanned by the kernel from different regions for the same single weight but for different neurons.

C. a single 2×2 feature map in a convolutional layer with 4 different neurons {G, H, I, J}

a 2×2 kernel containing 4 distinct sharing input weights {Wk1, Wk2, Wk3, Wk4} for the single 2×2 feature map in a convolutional layer.

a single shared bias weight W_0 for the single 2×2 feature map in a convolutional layer.

the first 4 inputs scanned by the 2×2 kernel for 4 distinct sharing weights {Wk1, Wk2, Wk3, Wk4} will be given to the set of weights of Neuron G.

the second 4 inputs scanned by the 2×2 kernel for 4 distinct sharing weights {Wk1, Wk2, Wk3, Wk4} will be given to the set of weights of Neuron H.

the third 4 inputs scanned by the 2×2 kernel for 4 distinct sharing weights {Wk1, Wk2, Wk3, Wk4} will be given to the set of weights of Neuron I.

the fourth 4 inputs scanned by the 2×2 kernel for 4 distinct sharing weights {Wk1, Wk2, Wk3, Wk4} will be given to the set of weights of Neuron J.

$\Rightarrow$ Gradient Descent Method: $\vec{W}' = \vec{W} + \eta \cdot (-\nabla e)$

$$\Rightarrow \text{Gradient Descent Method: } \begin{bmatrix} W_0' \\ W_{k1}' \\ W_{k2}' \\ W_{k3}' \\ W_{k4}' \end{bmatrix} = \begin{bmatrix} W_0 \\ W_{k1} \\ W_{k2} \\ W_{k3} \\ W_{k4} \end{bmatrix} + \eta \cdot \left(- \begin{bmatrix} \sum_{\{G,H,I,J\}} \left(\frac{\partial e}{\partial W_0}\right)_\xi \\ \sum_{\{G,H,I,J\}} \left(\frac{\partial e}{\partial W_{k1}}\right)_\xi \\ \sum_{\{G,H,I,J\}} \left(\frac{\partial e}{\partial W_{k2}}\right)_\xi \\ \sum_{\{G,H,I,J\}} \left(\frac{\partial e}{\partial W_{k3}}\right)_\xi \\ \sum_{\{G,H,I,J\}} \left(\frac{\partial e}{\partial W_{k4}}\right)_\xi \end{bmatrix} \right)$$

$$\Rightarrow \text{Gradient Descent Method: } \begin{bmatrix} W_0' \\ W_{k1}' \\ W_{k2}' \\ W_{k3}' \\ W_{k4}' \end{bmatrix} = \begin{bmatrix} W_0 \\ W_{k1} \\ W_{k2} \\ W_{k3} \\ W_{k4} \end{bmatrix} + \eta \cdot \left(- \begin{bmatrix} \left(\frac{\partial e}{\partial W_0}\right)_G + \left(\frac{\partial e}{\partial W_0}\right)_H + \left(\frac{\partial e}{\partial W_0}\right)_I + \left(\frac{\partial e}{\partial W_0}\right)_J \\ \left(\frac{\partial e}{\partial W_{k1}}\right)_G + \left(\frac{\partial e}{\partial W_{k1}}\right)_H + \left(\frac{\partial e}{\partial W_{k1}}\right)_I + \left(\frac{\partial e}{\partial W_{k1}}\right)_J \\ \left(\frac{\partial e}{\partial W_{k2}}\right)_G + \left(\frac{\partial e}{\partial W_{k2}}\right)_H + \left(\frac{\partial e}{\partial W_{k2}}\right)_I + \left(\frac{\partial e}{\partial W_{k2}}\right)_J \\ \left(\frac{\partial e}{\partial W_{k3}}\right)_G + \left(\frac{\partial e}{\partial W_{k3}}\right)_H + \left(\frac{\partial e}{\partial W_{k3}}\right)_I + \left(\frac{\partial e}{\partial W_{k3}}\right)_J \\ \left(\frac{\partial e}{\partial W_{k4}}\right)_G + \left(\frac{\partial e}{\partial W_{k4}}\right)_H + \left(\frac{\partial e}{\partial W_{k4}}\right)_I + \left(\frac{\partial e}{\partial W_{k4}}\right)_J \end{bmatrix} \right)$$

$$\Rightarrow \text{Gradient Descent Method: } \begin{bmatrix} W_0' \\ W_{k1}' \\ W_{k2}' \\ W_{k3}' \\ W_{k4}' \end{bmatrix} = \begin{bmatrix} W_0 \\ W_{k1} \\ W_{k2} \\ W_{k3} \\ W_{k4} \end{bmatrix} + \eta \cdot \left(- \begin{bmatrix} \left(\frac{\partial e}{\partial z_G} \cdot \frac{\partial z_G}{\partial W_0}\right)_G + \left(\frac{\partial e}{\partial z_H} \cdot \frac{\partial z_H}{\partial W_0}\right)_H + \left(\frac{\partial e}{\partial z_I} \cdot \frac{\partial z_I}{\partial W_0}\right)_I + \left(\frac{\partial e}{\partial z_J} \cdot \frac{\partial z_J}{\partial W_0}\right)_J \\ \left(\frac{\partial e}{\partial z_G} \cdot \frac{\partial z_G}{\partial W_{k1}}\right)_G + \left(\frac{\partial e}{\partial z_H} \cdot \frac{\partial z_H}{\partial W_{k1}}\right)_H + \left(\frac{\partial e}{\partial z_I} \cdot \frac{\partial z_I}{\partial W_{k1}}\right)_I + \left(\frac{\partial e}{\partial z_J} \cdot \frac{\partial z_J}{\partial W_{k1}}\right)_J \\ \left(\frac{\partial e}{\partial z_G} \cdot \frac{\partial z_G}{\partial W_{k2}}\right)_G + \left(\frac{\partial e}{\partial z_H} \cdot \frac{\partial z_H}{\partial W_{k2}}\right)_H + \left(\frac{\partial e}{\partial z_I} \cdot \frac{\partial z_I}{\partial W_{k2}}\right)_I + \left(\frac{\partial e}{\partial z_J} \cdot \frac{\partial z_J}{\partial W_{k2}}\right)_J \\ \left(\frac{\partial e}{\partial z_G} \cdot \frac{\partial z_G}{\partial W_{k3}}\right)_G + \left(\frac{\partial e}{\partial z_H} \cdot \frac{\partial z_H}{\partial W_{k3}}\right)_H + \left(\frac{\partial e}{\partial z_I} \cdot \frac{\partial z_I}{\partial W_{k3}}\right)_I + \left(\frac{\partial e}{\partial z_J} \cdot \frac{\partial z_J}{\partial W_{k3}}\right)_J \\ \left(\frac{\partial e}{\partial z_G} \cdot \frac{\partial z_G}{\partial W_{k4}}\right)_G + \left(\frac{\partial e}{\partial z_H} \cdot \frac{\partial z_H}{\partial W_{k4}}\right)_H + \left(\frac{\partial e}{\partial z_I} \cdot \frac{\partial z_I}{\partial W_{k4}}\right)_I + \left(\frac{\partial e}{\partial z_J} \cdot \frac{\partial z_J}{\partial W_{k4}}\right)_J \end{bmatrix} \right)$$

$$\Rightarrow \text{Gradient Descent Method: } \begin{bmatrix} W_0' \\ W_{k1}' \\ W_{k2}' \\ W_{k3}' \\ W_{k4}' \end{bmatrix} = \begin{bmatrix} W_0 \\ W_{k1} \\ W_{k2} \\ W_{k3} \\ W_{k4} \end{bmatrix} + \eta \cdot \left(- \begin{bmatrix} (E_{ror}G \cdot \text{the 1st input of } W_0) + (E_{ror}H \cdot \text{the 2nd input of } W_0) + (E_{ror}I \cdot \text{the 3rd input of } W_0) + (E_{ror}J \cdot \text{the 4th input of } W_0) \\ (E_{ror}G \cdot \text{the 1st input of } W_{k1}) + (E_{ror}H \cdot \text{the 2nd input of } W_{k1}) + (E_{ror}I \cdot \text{the 3rd input of } W_{k1}) + (E_{ror}J \cdot \text{the 4th input of } W_{k1}) \\ (E_{ror}G \cdot \text{the 1st input of } W_{k2}) + (E_{ror}H \cdot \text{the 2nd input of } W_{k2}) + (E_{ror}I \cdot \text{the 3rd input of } W_{k2}) + (E_{ror}J \cdot \text{the 4th input of } W_{k2}) \\ (E_{ror}G \cdot \text{the 1st input of } W_{k3}) + (E_{ror}H \cdot \text{the 2nd input of } W_{k3}) + (E_{ror}I \cdot \text{the 3rd input of } W_{k3}) + (E_{ror}J \cdot \text{the 4th input of } W_{k3}) \\ (E_{ror}G \cdot \text{the 1st input of } W_{k4}) + (E_{ror}H \cdot \text{the 2nd input of } W_{k4}) + (E_{ror}I \cdot \text{the 3rd input of } W_{k4}) + (E_{ror}J \cdot \text{the 4th input of } W_{k4}) \end{bmatrix} \right)$$